Project: Analyzing Transportation Trends in Europe Using Python

1. Project Overview

Title: Analyzing Transportation Trends in Europe Using Python

Objective:

To analyze transportation patterns and trends across European countries and regions using real-world datasets. The project explores how passenger and freight transportation have evolved over time and how transportation modes, including road, rail, air, maritime transport, public transportation, cycling, and micromobility, differ across geographical areas.

The project may also investigate the relationship between transportation and sustainability, including environmental impacts, emissions, and sustainable mobility, as well as the relationship between transportation accessibility and housing prices, rents, affordability, and urban development.

Students will use Python to collect, clean, analyze, and visualize transportation data and develop a Flask-based web application to present their findings interactively.

Tools and Technologies:

Python for data analysis and visualization, Flask for web application development, and libraries such as Pandas, NumPy, Matplotlib, Plotly, and Seaborn.

2. Key Learning Outcomes

- **Python Programming Proficiency:** Learn to write structured Python programs using functions, classes, and libraries for data analysis.
- **Data Collection and Integration:** Gain experience accessing, importing, and combining datasets using APIs or CSV/Excel files.
- **Data Cleaning and Preprocessing:** Develop skills to handle missing data, transform variables, and prepare datasets for analysis.
- **Exploratory Data Analysis (EDA):** Apply Python to summarize, group, and analyze transportation data across countries, regions, transportation modes, and years.
- **Data Visualization and Storytelling:** Use libraries such as Matplotlib, Plotly, and Seaborn to create clear, interactive, and meaningful visual representations of transportation patterns.
- **Web App Development:** Build a simple Flask-based web application that allows users to interact with the data through filters and charts.

- **Data-Driven Decision-Making:** Investigate transportation trends and relationships with environmental, demographic, economic, and housing indicators, where relevant.

3. Project Steps

The project steps should follow an adapted version of the CRISP-DM/POST-DS methodology. A suggested structure is provided below.

Step 1: Understanding the Problem and the Data

Access transportation datasets through APIs or downloads from sources such as:

- Eurostat Transport Database — passenger transport, freight transport, infrastructure, road safety, and other transportation indicators.
- Eurostat Data Browser — transportation and socioeconomic datasets, including population, housing, and regional indicators.
- European Commission — Mobility and Transport — information on European transport systems and policies.
- European Environment Agency — environmental indicators, including transport-related emissions.
- European Union Open Data Portal — additional datasets from European institutions and public authorities.
- OpenStreetMap — geographic information on roads, cycleways, public transport infrastructure, and other points of interest.
- National and municipal open-data portals — potentially useful sources for public transport, shared bicycles, e-scooters, cycling infrastructure, housing, and urban mobility.

Explore the data structure, identify key variables (e.g., country, region, year, transport mode, passenger numbers, freight volumes, infrastructure, emissions, housing prices, or population), and check for missing or inconsistent values.

Each group should select a specific topic and formulate one or more research questions that can be answered using the available data.

Possible research questions include:

- How have passenger transportation patterns evolved across European countries?
- How does the distribution of passenger transport between road, rail, and air differ across countries?
- How has micromobility, including shared bicycles and electric scooters, evolved in selected European cities?
- What relationships exist between cycling infrastructure and micromobility usage?
- How do transportation modes differ in their environmental impact?
- What relationships exist between public transportation usage and transport-related emissions?
- Is accessibility to public transportation associated with housing prices or rents?

- How do housing prices vary with proximity to railway stations or metro stations?
- What relationships exist between public transportation availability, population density, and housing affordability?
- How do transportation accessibility and environmental sustainability vary across urban areas?

The questions are examples, not mandatory research objectives. Each group should focus on a manageable research problem consistent with its data sources.

Step 2: Data Preparation

Clean and organize the data for analysis:

- Handle missing values, duplicates, and inconsistent observations.
- Standardize country codes, regional identifiers, dates, and units of measurement.
- Transform variables where necessary, including converting units and calculating rates.
- Merge datasets when necessary, ensuring that geographic identifiers, time periods, and measurement units are compatible.
- Label geographic regions correctly using appropriate regional identifiers, including NUTS codes when applicable.
- Create summary tables by country, region, transportation mode, or year.
- Develop relevant indicators, such as percentage changes, modal shares, transport emissions per capita, or housing prices relative to income.

For projects combining transportation and housing data, students should ensure that the datasets refer to compatible geographical areas and time periods. When datasets are available at different spatial resolutions, the limitations of combining them must be explained.

Students should document their data preparation decisions and explain how these decisions affect the subsequent analysis.

Step 3: Data Analysis

Perform exploratory analysis using Pandas and NumPy.

Depending on the selected research question, students may:

- Identify overall transportation trends over time.
- Compare transportation modes, such as road, rail, air, maritime transport, public transport, cycling, and micromobility.
- Evaluate which countries or regions have the highest and lowest values for selected transportation indicators.
- Analyze differences between passenger and freight transport.
- Investigate relationships between transportation indicators and contextual variables, such as GDP, population, population density, or urbanization.

- Examine transportation sustainability through indicators such as transport-related emissions or the modal share of public transport and active mobility.
- Investigate relationships between transportation accessibility and housing prices, rents, or affordability.
- Explore differences in micromobility adoption and cycling infrastructure across cities.
- Calculate descriptive statistics, growth rates, proportions, and other relevant indicators.
- Identify unusual observations and discuss possible explanations supported by evidence.

Students may use correlation analysis and simple regression models when these methods are appropriate for their research questions and data.

The analysis should be guided by the selected research questions rather than consisting only of descriptive charts. Statistical associations must not automatically be interpreted as causal relationships.

Step 4: Data Visualization

Create visualizations to make findings clear and engaging:

- **Line charts:** Show changes in transportation indicators, micromobility usage, emissions, or housing prices over time.
- **Bar charts:** Compare countries, regions, transportation modes, or selected indicators.
- **Stacked bar charts:** Compare the composition of passenger or freight transportation across modes.
- **Heatmaps:** Visualize patterns across countries and years or correlations between suitable numerical variables.
- **Interactive maps (Plotly):** Display transportation indicators, cycling infrastructure, mobility patterns, or housing-related indicators across geographical areas, where suitable geographic data are available.
- **Scatter plots:** Investigate relationships between transportation accessibility, emissions, housing prices, population density, or other relevant variables.
- **Animations (optional):** Show how transportation or housing indicators evolve over time.

Each visualization should have a clear purpose, appropriate labels, readable scales, and a short interpretation explaining the main findings.

Students should select chart types according to the characteristics of their data and research questions. Interactive maps and spatial visualizations are particularly relevant when the analysis involves urban mobility, micromobility, or transportation accessibility.

Step 5: Building a Flask Web App

Develop a basic web application that allows users to interact with the data through filters and charts.

The application should allow users to:

- Select a country or region.
- Select a transportation mode or indicator.
- Choose a time period.
- View relevant charts and key statistics.
- Compare countries, regions, or transportation modes.
- Explore trends interactively through embedded visualizations.

Depending on the project topic, the application may also allow users to:

- Explore micromobility and cycling indicators.
- Compare transportation sustainability indicators.
- Investigate relationships between transport accessibility and housing indicators.
- Display interactive maps of transportation infrastructure or relevant geographical patterns.

Include a simple dashboard that highlights the main insights from the analysis.

The application should use the group's prepared dataset and analytical outputs rather than merely displaying static screenshots of charts.

Step 6: Deployment

Deploy the Flask application using a suitable hosting platform, such as PythonAnywhere or another platform that supports Flask applications.

Before submission, verify that the application runs correctly, its principal features work as expected, and the submitted link is accessible.

If deployment is not possible because of technical constraints, the group should provide clear instructions for running the application locally and explain the limitation.

4. Extensions and Optional Features

Groups may extend their projects with additional functionality, provided that the core requirements are completed first.

- Add filters to compare countries, regions, transportation modes, and time periods.
- Include explanatory texts, tooltips, and definitions of the indicators used.
- Add a download option for filtered data or charts.
- Integrate additional datasets, such as population, GDP, housing, or environmental indicators.
- Develop interactive maps showing transportation infrastructure, micromobility services, or accessibility indicators.
- Compare housing prices or rents across areas with different levels of transportation accessibility.
- Investigate the relationship between cycling infrastructure and micromobility usage.
- Compare environmental indicators across transportation modes.

- Implement simple forecasting models to estimate future trends in a selected transportation or housing indicator, provided that sufficient historical data are available.

Additional analyses should be justified by the research question and supported by the available data.

5. The Final Report Structure and Format

Authors and Affiliation: ISEG or another relevant institution.

Abstract

The abstract should summarize the project, including the problem investigated, objectives, data sources, methodology, principal findings, and conclusions.

1. Introduction

Provide an overview of the project objectives, motivation, and relevance of studying transportation trends in Europe.
Introduce the selected transportation topic, explain the research problem, and state the research questions.
Depending on the project, discuss the relevance of mobility, micromobility, transportation sustainability, environmental impacts, accessibility, or housing.
Briefly explain how Python and data science can contribute to understanding transportation patterns and supporting evidence-based decision-making.

2. Theoretical Framework

Summarize the concepts relevant to the selected research topic.
Depending on the project, these may include transportation modes, passenger mobility, freight transport, micromobility, sustainable transportation, accessibility, environmental impacts, housing affordability, or urban development.
Explain the relevance of data-driven decision-making for the selected problem.
Present the CRISP-DM/POST-DS methodology as the conceptual foundation guiding the project [1].
The theoretical framework should focus on concepts directly related to the research questions rather than providing a general description of all transportation topics.

3. Methodological Approach

Explain how the CRISP-DM/POST-DS phases were applied.
Describe:

- The research questions and analytical objectives.
- The selected datasets and their sources.
- The variables and indicators used.
- The data collection and preparation procedures.
- The analytical methods and visualizations applied.
- The development and implementation of the Flask application.

Explain the main methodological decisions and any limitations affecting the analysis.

4. Results (Description of Steps and Their Outcomes)

Describe and illustrate the main analytical steps and their outputs.

Include:

- **Data collection and preparation:** Describe the datasets, variables, cleaning procedures, transformations, and resulting analytical dataset.
- **Exploratory analysis:** Present descriptive statistics and the main patterns identified.
- **Visualization of trends and comparisons:** Include relevant time-series charts, bar charts, maps, scatter plots, or other visualizations.
- **Interpretation:** Discuss the findings in relation to the research questions, including differences across countries, regions, transportation modes, or periods.
- **Flask application:** Include screenshots and an interactive description of the main features.

Depending on the selected topic, the results may address transportation and micromobility patterns, sustainability indicators, environmental impacts, transportation accessibility, housing prices, or relationships between these dimensions.
Interpret the results critically. Distinguish between observed patterns, statistical associations, and causal explanations. Do not claim that one variable causes another solely because a correlation has been observed.

5. Conclusions

Summarize the main insights gained and explain how the findings address the research questions.
Discuss the most relevant transportation trends, comparisons, and differences identified in the data.
Depending on the project, reflect on implications for mobility, micromobility, environmental sustainability, housing affordability, or urban development.
Discuss the practical value of the analysis, the limitations of the datasets and methods, and possible extensions, such as forecasting, additional data integration, or more detailed spatial analysis.

References

Include all references to datasets, documentation, software, and academic sources used in the project.
The following methodological reference is mandatory:
[1] C. J. Costa and J. T. Aparicio, "POST-DS: A Methodology to Boost Data Science," 2020 15th Iberian Conference on Information Systems and Technologies (CISTI), Seville, Spain, 2020, pp. 1–6, IEEE, doi: 10.23919/CISTI49556.2020.9140932.
Include references to the Eurostat transportation datasets and any other data sources, academic publications, or technical documentation used.

6. Submission

Create a folder named `ProjectXX`, where `XX` is your group number in your channel, and upload:

- All Python and/or Jupyter Notebook files.
- The source code for the Flask application.
- The data files used, or clear instructions and scripts for retrieving the data.
- A short report in PDF **and** DOC/DOCX format. Maximum 10 pages font 10 Times new roman..
- The link to the deployed Flask application, or instructions for running it locally if deployment is not possible.

The project files should be organized clearly so that another person can understand the workflow and reproduce the main results.

7. Evaluation Criteria

- **Code Quality:** Clear structure, appropriate use of functions and classes, readability, useful comments, and efficient use of libraries.
- **Data Collection and Preparation:** Appropriate data sources, correct data integration, documented cleaning procedures, and consistency of the resulting dataset.
- **Data Visualization:** Clarity, appropriate chart selection, effective design, and useful interactive features.
- **Analysis Quality:** Relevance of the research questions, correctness of the analysis, depth of insights, and logical interpretation of the findings.
- **Web App Functionality:** Simplicity, usability, correct operation, and meaningful interaction with the data.
- **Report and Presentation:** Organization, clarity, completeness, critical discussion, and appropriate references. The methodological reference presented above is mandatory.
- **Attendance and Participation:** All students must attend class and present their project.
- **Peer Evaluation and Engagement:** Students must attend their colleagues' presentations, not only as a matter of academic respect but also to participate in their evaluation.